

HYDROGENATED NBR (HNBR) SYNTHESIS AND PROCESS PATENTS LANDSCAPE

1. ABSTRACT

This report synthesizes seven primary patents relevant to hydrogenated nitrile rubber (HNBR), focusing on routes that either directly hydrogenate NBR or combine hydrogenation with molecular-weight reduction to improve processability. The target material is hydrogenated nitrile rubber derived from nitrile rubber precursors containing conjugated diene and α,β -unsaturated nitrile units, with emphasis on low residual unsaturation, improved heat/oil/ozone resistance, and controlled molecular weight or Mooney viscosity. Across the patent set, two major process families emerge. First, several patents (US7919563B2, US7772328B2, EP2473281B1, US9150669B2) disclose metathesis-based depolymerization of NBR/HNBR precursors using ruthenium- or osmium-based catalysts, often with co-olefins such as ethylene or 1-hexene, to reduce molecular weight and narrow polydispersity. EP2473281B1 and US9150669B2 are especially notable for simultaneous metathesis and hydrogenation in the presence of hydrogen, enabling one-step formation of low-MW HNBR with low residual double bond content. Second, US10759885B2 and US10717906B2 are not direct HNBR synthesis patents but are relevant to hydrogenated elastomer technology: US10759885B2 discloses hydrogenation of branched conjugated diene copolymers (e.g., myrcene/farnesene/styrene systems) to improve tire performance, while US10717906B2 uses partially hydrogenated hydrocarbon resins in adhesive formulations, illustrating broader formulation trends around hydrogenated unsaturation for stability and performance. US7846995B2 addresses downstream purification by removing ruthenium residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins, an important process-enabling step for catalyst-sensitive HNBR production. Overall, the landscape shows a progression from conventional two-step metathesis-then-hydrogenation toward single-step hydrogenation/metathesis, with strong attention to catalyst choice, co-olefin selection, residual double bond control, and post-reaction metal removal.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7919563B2

- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for preparing low-molecular-weight hydrogenated nitrile rubber by subjecting nitrile rubber to a metathesis reaction in the presence of ruthenium- or osmium-based catalysts, optionally with a co-olefin and solvent. The process is aimed at reducing molecular weight and narrowing molecular weight distribution while retaining the HNBR microstructure. - Molecular weight (MW): 30,000 to 250,000 — Claim 1 - Mooney viscosity (ML 1+4 @ 100°C): 3 to 50 — Claim 1 - Polydispersity index (MWD): less than 2.5 — Claim 1 - Cement concentration: 15 wt.-% solid — Example 1 - Rubber mass: 200 g — Example 1 - MCB mass: 1133 g — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Degassing cycles: 3 — Example 1 - Rubber mass: 75 g — Example 3 - MCB mass: 1175 g — Example 3 - Cement concentration: 6 wt.-% solid — Example 3 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing cycles: 3 — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin concentration: 500 psi — Table 1 - Co-olefin: ethylene — Table 1 - Co-olefin: 1-hexene — Table 1 - Mn: 27 kg/mol — Example 4 discussion - Mw: 54 kg/mol — Example 4 discussion - Starting polymer Mn: 85 kg/mol — Example 4 discussion - Starting polymer Mw: 296 kg/mol — Example 4 discussion - PDI: 2.0 — Example 4 discussion - PDI: 1.90 — Table 2 - Mooney viscosity: 73 — Table 2 (Therban® A3407 substrate) - Mooney viscosity: 43 — Table 2 (Experiment 1) - Mooney viscosity: 28 — Table 2 (Experiment 2) - Mooney viscosity: 3 — Table 2 (Experiment 3) - Mooney viscosity: 31 — Table 2 (Experiment 4)

Relevant Experimental Evidence - Example 1: Ethylene-assisted metathesis of NBR cement at 15 wt.-% solids under 6.9×10^5 Pa ethylene and full agitation was demonstrated. - Example 2: A second ethylene-assisted run repeated the same cement and pressure conditions, supporting reproducibility. - Example 3: Lower-solids cement (6 wt.-%) was processed under the same ethylene pressure to show the method works at different polymer concentrations. - Example 4: 1-hexene was used as co-olefin at 80°C with 0.25 phr catalyst, giving substantial molecular-weight reduction and lower Mooney viscosity. - Example 4: Product data showed Mn 27 kg/mol, Mw 54 kg/mol, and PDI 2.0, confirming controlled depolymerization.

Technical Relevance This patent is directly relevant because it targets low-MW HNBR and demonstrates how metathesis conditions can tune viscosity and polydispersity. It provides a practical route for improving HNBR processability while preserving the nitrile rubber backbone chemistry.

Patent 2

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses the same low-molecular-weight HNBR concept as US7919563B2, using metathesis of nitrile rubber with ruthenium/osmium catalysts and optional co-olefins to reduce molecular weight and Mooney viscosity. The examples again emphasize ethylene or 1-hexene as co-olefin choices and show the effect of catalyst loading and reactor temperature on product properties. - Molecular weight (MW): 30,000 to 250,000 — Claim 1 - Mooney viscosity (ML 1+4 @100°C): less than 10 — Claim 1 - Polydispersity index (MWD): less than 2.5 — Claim 1 - Molecular weight (Mw): 40,000 to 220,000 — Claim 2 - Polydispersity index: less than 2.3 — Claim 3 - Cement concentration: 15 wt.-% solid — Example 1 - Rubber mass: 200 g — Example 1 - MCB mass: 1133 g — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Rubber mass: 75 g — Example 3 - MCB mass: 1175 g — Example 3 - Cement concentration: 6 wt.-% solid — Example 3 - 1-hexene amount: 150 g — Example 4 - Nitrogen degassing cycles: 3 — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Co-olefin concentration: 500 psi — Table 1 - Mn: 27 kg/mol — Example 4 discussion - Mw: 54 kg/mol — Example 4 discussion - Starting polymer Mn: 85 kg/mol — Example 4 discussion - Starting polymer Mw: 296 kg/mol — Example 4 discussion - PDI: 2.0 — Example 4 discussion - PDI: 1.90 — Table 2 - Mooney viscosity: 73 — Table 2 (substrate) - Mooney viscosity: 43 — Table 2 (Experiment 1) - Mooney viscosity: 28 — Table 2 (Experiment 2) - Mooney viscosity: 3 — Table 2 (Experiment 3) - Mooney viscosity: 31 — Table 2 (Experiment 4)

Relevant Experimental Evidence - Example 1: Ethylene degassing at 6.9×10^5 Pa was used to initiate metathesis of a 15 wt.-% rubber cement. - Example 2: The same ethylene-based setup was repeated to confirm the process. - Example 3: A lower-solids cement was processed under identical ethylene pressure, showing flexibility in feed concentration. - Example 4: 1-hexene co-olefin at 80°C with 0.25 phr catalyst produced low-MW HNBR with reduced Mooney viscosity. - Example 4: The product reached PDI 1.90 and Mooney viscosity 3, demonstrating strong depolymerization control.

Technical Relevance This patent is highly relevant because it explicitly targets low-MW HNBR and quantifies the resulting viscosity and molecular-weight improvements. It reinforces the importance of co-olefin-assisted metathesis for processable HNBR grades.

Patent 3

- Patent Number: EP2473281B1

- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneously subjecting nitrile rubber to metathesis and hydrogenation in the presence of hydrogen and a ruthenium- or osmium-based catalyst. The process is positioned as an alternative to sequential metathesis followed by hydrogenation, with the goal of lowering residual double bonds, molecular weight, and Mooney viscosity in a single operation. - Molecular weight (Mw): 20,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - Polydispersity index (MWD): less than 3.0 — Description - Molecular weight (Mw): 20,000 to 250,000 — Claim 1 - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Claim 1 - Polydispersity index (MWD): less than 3.0 — Claim 1 - Mooney viscosity (ML 1+4 @100°C): less than 10 — Claim 2 - Hydrogenation ratio: 10 to 90% — Description - Hydrogenation ratio: 30 to 70% — Description - Tg: -80°C to 110°C — Description - Tg: -70°C to 70°C — Description - Tg: -30°C to 30°C — Description - Mw: 1,000 or more — Description - Mw: 3,000,000 or less — Description - Mn: 3,000 or more — Description - Mn: 3,000,000 or less — Description - Mw/Mn: 20.0 or less — Description - Mw/Mn: 10.0 or less — Description - Conjugated diene content: 1 to 18% — Description - Mooney viscosity of commercial HNBR: 55 — Description - RDB of commercial HNBR: 18% — Description

Relevant Experimental Evidence - Example a: The patent discussion contrasts prior metathesis catalysts and explains why simultaneous hydrogenation/metathesis is needed. - Example a: It identifies Grubbs-type catalysts and related Ru systems as prior art but notes limitations for direct hydrogenation. - Example a: The disclosure emphasizes that the inventive process can produce low-RDB HNBR in a single step. - Example a: The patent frames the process as an alternative to sequential metathesis then hydrogenation, reducing process complexity.

Technical Relevance This is one of the most directly relevant patents because it claims simultaneous metathesis and hydrogenation of NBR to HNBR. It is central to the target synthesis problem of controlling residual unsaturation and molecular weight in a single step.

Patent 4

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent also discloses a single-step process for producing low-molecular-weight HNBR by contacting nitrile rubber with a ruthenium- or osmium-based catalyst in the presence of hydrogen, optionally with a co-olefin. The examples compare different hydrogenation catalysts and show that the inventive catalyst system can simultaneously reduce residual double bonds and molecular weight. - Molecular weight (Mw): 20,000 to 250,000 — Description - Mooney viscosity (ML 1+4 @ 100°C): 1 to 50 — Description - Polydispersity index (MWD): less than 3.0 — Description - Catalyst amount: 0.041 phr — Example 1 - Catalyst amount: 0.045 phr — Example 1 - Catalyst amount: 0.060 phr — Example 1 - Metal content: 65.6 ppm — Example 1 - Co-catalyst: triphenylphosphine — Example 1 - Co-catalyst amount: 1.0 phr — Example 1 - Reaction time: 0 min — Example 1 - Reaction time: 60 min — Example 1 - Reaction time: 120 min — Example 1 - Reaction time: 180 min — Example 1 - Reaction time: 240 min — Example 1 - RDB: 100% — Example 1, 0 min - RDB: 2.5% — Example 1, 60 min (HNBR 1) - RDB: 7.0% — Example 1, 60 min (HNBR 2) - RDB: 15.4% — Example 1, 60 min (HNBR 3) - RDB: 0.6% — Example 1, 120 min (HNBR 1) - RDB: 4.5% — Example 1, 120 min (HNBR 2) - RDB: 3.1% — Example 1, 120 min (HNBR 3) - RDB: 0.5% — Example 1, 180 min (HNBR 1) - RDB: 3.9% — Example 1, 180 min (HNBR 2) - RDB: 1.3% — Example 1, 180 min (HNBR 3) - RDB: 0.4% — Example 1, 240 min (HNBR 1) - RDB: 3.5% — Example 1, 240 min (HNBR 2) - RDB: 0.7% — Example 1, 240 min (HNBR 3) - Mw: 113371 g/mol — Example 1, HNBR 1 - Mn: 48352 g/mol — Example 1, HNBR 1 - PDI: 2.34 — Example 1, HNBR 1 - Mooney viscosity: 33.3 — Example 1, HNBR 1 - Mw: 133758 g/mol — Example 1, HNBR 2 - Mn: 54595 g/mol — Example 1, HNBR 2 - PDI: 2.45 — Example 1, HNBR 2 - Mooney viscosity: 41.3 — Example 1, HNBR 2 - Mw: 195304 g/mol — Example 1, HNBR 3 - Mn: 64130 g/mol — Example 1, HNBR 3 - PDI: 3.05 — Example 1, HNBR 3 - Mooney viscosity: 69.3 — Example 1, HNBR 3

Relevant Experimental Evidence - Example 1: Compared Hoveyda, Arlt, and Wilkinson-type hydrogenation catalysts at similar metal content (65.6 ppm). - Example 1: The inventive catalysts achieved faster RDB reduction than the industrial norm. - Example 1: After 240 min, HNBR 1 reached 0.4% RDB and Mooney 33.3, while HNBR 3 remained at 0.7% RDB and Mooney 69.3. - Example 1: The inventive systems also reduced molecular weight relative to the industrial benchmark. - Example 1: The data support simultaneous hydrogenation/metathesis as a route to lower-viscosity HNBR.

Technical Relevance This patent is highly relevant because it demonstrates the target one-step hydrogenation/metathesis concept with quantitative catalyst and product data. It provides a practical benchmark for catalyst selection and expected reductions in RDB and Mooney viscosity.

Patent 5

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber

- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not synthesize HNBR itself; instead, it discloses a post-treatment process for removing ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins. The process is important for cleaning up metathesis-derived HNBR solutions and lowering residual metal content in the final polymer. - Ruthenium in solution: 5 to 1000 ppm — Claim 2 - Ruthenium in solution: 5 to 500 ppm — Claim 3 - Polymer concentration: 0.5 to 30 wt.% — Claim 4 - Resin particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Treatment temperature: ca. 100°C — Example 2A - Treatment time: 66 hours — Example 2A - Initial Ru content: 15 ppm — Table 2 - Ru content after inventive process: 10 ppm — Table 2 - Ru removal: 33% — Table 2 - Resin mass: 0.5 g — Example 2A - Rubber solution mass: 180 g — Example 2A - Drying temperature: 60°C — Example 2A

Relevant Experimental Evidence - Example 2A: A nitrile rubber solution was treated with 0.5 g of Lewatit® MonoPlus MP 500 at about 100°C for 66 hours. - Example 2A: The treatment reduced Ru content from 15 ppm to 10 ppm. - Example 2A: The process achieved 33% ruthenium removal from the rubber solution. - Example 2B: An untreated control sample was used to establish the baseline Ru content. - Example 2: The data show that ion-exchange cleanup can meaningfully reduce catalyst residue in hydrogenated nitrile rubber.

Technical Relevance This patent is relevant because catalyst residue removal is a critical downstream step in HNBR manufacture, especially for ruthenium-based metathesis/hydrogenation systems. It addresses product purity and process robustness rather than polymer formation itself.

Patent 6

- Patent Number: US10759885B2
- Patent Title: Hydrogenated branched conjugated diene copolymer, rubber composition and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of hydrogenated branched conjugated diene copolymers by first copolymerizing branched

conjugated diene monomers with vinyl compounds and then hydrogenating the resulting copolymer. Although not an NBR patent, it is relevant as a hydrogenation-based elastomer synthesis route that emphasizes control of hydrogenation ratio, molecular weight, and processability. - Copolymerization ratio of branched conjugated diene compound: 1 to 99 wt.% — Claim 1 - Copolymerization ratio of vinyl compound: 99 to 1 wt.% — Claim 1 - Hydrogenation ratio: 10 to 90% — Description - Hydrogenation ratio: 30 to 70% — Description - Mw: 1,000 or more — Description - Mw: 3,000,000 or less — Description - Mw: 3,000 or more — Description - Mw: 200,000 or less — Description - Mn: 3,000 or more — Description - Mn: 3,000,000 or less — Description - Mw/Mn: 20.0 or less — Description - Mw/Mn: 10.0 or less — Description - Tg: -80°C to 110°C — Description - Tg: -70°C to 70°C — Description - Tg: -30°C to 30°C — Description - 2000 ml hexane — Preparation Example 1 - 110 g butadiene — Preparation Example 1 - 90 g styrene — Preparation Example 1 - 0.22 mmol TMEDA — Preparation Example 1 - 35 mmol n-BuLi — Preparation Example 1 - Polymerization time: 5 hours — Preparation Example 1 - Polymerization temperature: 50°C — Preparation Example 1 - 1.15 ml of 1M isopropanol/hexane — Preparation Example 1 - 200 g Copolymer 1 — Preparation Example 1 - 150°C kneading time: 5 min — Preparation Example 1-2 - 170°C kneading time: 12 min — Preparation Example 1-2 - Hydrogenation catalyst: 10% palladium carbon — Preparation Example 2 - THF: 300 g — Preparation Example 2 - Hydrogen pressure: 5.0 kgf/cm² — Preparation Example 2 - Hydrogenation temperature: 80°C — Preparation Example 2 - Hydrogenation degree: 50% — Preparation Example 2 - 500 ml hexane — Preparation Example 3 - 46 g THF — Preparation Example 3 - 45 mmol n-BuLi — Preparation Example 3 - 150 g isoprene — Preparation Example 3 - 125 g styrene — Preparation Example 3 - Dropwise addition time: 2 hours — Preparation Example 3 - 20 ml of 2M isopropanol/hexane — Preparation Example 3 - 150 g myrcene — Preparation Example 5 - 40 mmol n-BuLi — Preparation Example 5 - 150 g farnesene — Preparation Example 5 - 95 g butadiene — Preparation Example 7 - 55 g myrcene — Preparation Example 7 - 125 g styrene — Preparation Example 7 - 55 g farnesene — Preparation Example 17 - 95 g butadiene — Preparation Example 17

Relevant Experimental Evidence - Preparation Example 1: Butadiene/styrene copolymer was synthesized in hexane with TMEDA and n-BuLi at 50°C, then used in tire-compound preparation. - Preparation Example 2: Hydrogenation of Copolymer 1 with 10% Pd/C at 5.0 kgf/cm² H₂ and 80°C gave Copolymer 2 with 50% hydrogenation. - Preparation Example 3: Isoprene/styrene copolymerization under n-BuLi initiation produced a nearly fully converted copolymer. - Preparation Example 5: Farnesene/styrene copolymerization was demonstrated as a branched diene precursor route. - Preparation Example 7: A ternary butadiene/myrcene/styrene copolymer was synthesized, showing the branched-diene design space before hydrogenation.

Technical Relevance This patent is indirectly relevant because it shows how hydrogenation is used to tune elastomer properties and processability in branched diene copolymers. The hydrogenation conditions and property targets provide useful context for broader hydrogenated elastomer synthesis, though not specifically HNBR.

Patent 7

- Patent Number: US10717906B2
- Patent Title: Self-adhesive compound and use thereof in an adhesive tape
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not an HNBR synthesis patent; it discloses self-adhesive compositions containing polybutadiene block copolymers and partially hydrogenated hydrocarbon resins. It is relevant as a formulation example showing how partial hydrogenation is used to improve stability, transparency, and ozone resistance in elastomer-based products. - Elastomer fraction: at least 40 wt.% — Claim 1 - Partly hydrogenated hydrocarbon resin softening temperature: at least 90°C — Claim 1 - Plasticizing resin fraction: not more than 6 wt.% — Claim 1 - Plasticizing resin softening temperature: <25°C — Claim 1 - $\tan \delta$ at 25°C and 10 rad/s: at least 0.20 — Claim 1 - $\tan \delta$ at 25°C and 100 rad/s: at least 0.40 — Claim 1 - $\tan \delta$ at 25°C and 10 rad/s: not more than 0.40 — Claim 1 - $\tan \delta$ at 25°C and 100 rad/s: not more than 1.00 — Claim 1 - Kraton D1165 PT: 24.00 wt.% — Example 1 - Dercolyte A115: 47.00 wt.% — Example 1 - Wingtack 10: 3.50 wt.% — Example 1 - Kraton D1102 CS: 24.00 wt.% — Example 1 - Kraton D1102 CS: 39.00 wt.% — Example 2 - Dercolyte A115: 45.00 wt.% — Example 2 - Wingtack 10: 4.50 wt.% — Example 2 - Kraton D1118 CS: 10.00 wt.% — Example 2 - Kraton D1102 CS: 38.00 wt.% — Example 3 - Sukorez SU110: 23.50 wt.% — Example 3 - Sukorez SU100: 23.50 wt.% — Example 3 - Wingtack 10: 3.50 wt.% — Example 3 - Kraton D1102 CS: 39.00 wt.% — Example 4 - Regalite R1100: 30.00 wt.% — Example 4 - Foral AX: 16.00 wt.% — Example 4 - Kraton D1102 CS: 38.00 wt.% — Example 5 - Regalite R7100: 32.00 wt.% — Example 5 - Piccolyte A115: 15.00 wt.% — Example 5 - Kraton D1102 CS: 38.00 wt.% — Example 6 - Regalite R7100: 37.00 wt.% — Example 6 - Regalite S5090: 10.00 wt.% — Example 6 - Kraton D1102 CS: 30.00 wt.% — Example 7 - Ragalite R7100: 47.00 wt.% — Example 7 - Arkon M90: 13.00 wt.% — Example 7 - Kraton D1118 CS: 5.00 wt.% — Example 7 - Kraton D1118 CS: 98.50 wt.% — Example 8 - Kraton D1165 PT: 98.50 wt.% — Example 9 - Kraton D1102 CS: 98.50 wt.% — Example 10 - $\tan \delta$ 25°C, 10 rad/s: 0.73 — Example 1 - $\tan \delta$ 25°C, 100 rad/s: 1.55 — Example 1 - $\tan \delta$ 25°C, 10 rad/s: 0.49 — Example 2 - $\tan \delta$ 25°C, 100 rad/s: 1.15 — Example 2 - $\tan \delta$ 25°C, 10 rad/s: 0.25 — Example 3 - $\tan \delta$ 25°C, 100 rad/s: 0.53 — Example 3 - $\tan \delta$ 25°C, 10 rad/s: 0.17 — Example 4 - $\tan \delta$ 25°C, 100 rad/s: 0.30 — Example 4 - $\tan \delta$ 25°C, 10 rad/s: 0.26 — Example 5 - $\tan \delta$ 25°C, 100 rad/s: 0.65 — Example 5 - $\tan \delta$ 25°C, 10 rad/s: 0.24 — Example 6 - $\tan \delta$ 25°C, 100 rad/s: 0.45 — Example 6 - $\tan \delta$ 25°C, 10 rad/s: 0.22 — Example 7 - $\tan \delta$ 25°C, 100 rad/s: 0.42 — Example 7 - $\tan \delta$ 25°C, 10 rad/s: 0.11 — Example 8 - $\tan \delta$ 25°C, 100 rad/s: 0.11 — Example 8 - $\tan \delta$ 25°C, 10 rad/s: 0.08 — Example 9 - $\tan \delta$ 25°C, 100 rad/s: 0.08 — Example 9 - $\tan \delta$ 25°C, 10 rad/s: 0.06 — Example 10 - $\tan \delta$ 25°C, 100 rad/s: 0.06 — Example 10

Relevant Experimental Evidence - Example 1: A comparative adhesive formulation used nonhydrogenated resin and showed strong tack but ozone damage. - Example 3: Fully hydrogenated resins improved ozone resistance but altered viscoelastic balance. - Example 5: An inventive formulation with partially hydrogenated resin and nonhydrogenated resin achieved improved balance of properties. - Example 6: A second inventive formulation with two partially hydrogenated resins showed good ozone resistance and acceptable $\tan \delta$. - Example 10: A formulation lacking tackifying resins performed poorly, underscoring the importance of resin hydrogenation state and blend design.

Technical Relevance This patent is only indirectly relevant to HNBR synthesis, but it illustrates how partial hydrogenation is used to tune elastomeric performance and ozone stability. It supports the broader materials-design principle that hydrogenation level strongly affects viscoelastic and durability properties.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US7772328B2 both use metathesis-based molecular-weight reduction of nitrile rubber with ruthenium/osmium catalysts, optional co-olefins, and elevated temperature; both emphasize low MW and low Mooney HNBR, but US7772328B2 explicitly claims Mooney <10 and Mw 40,000 to 220,000 in dependent claims.
- EP2473281B1 and US9150669B2 are the key single-step HNBR patents: both run nitrile rubber in hydrogen with Ru/Os catalysts and optional co-olefin, aiming for simultaneous metathesis/hydrogenation rather than sequential processing.
- US9150669B2 provides the most explicit comparative catalyst data, including Hoveyda, Arlt, and Wilkinson-type systems, 65.6 ppm metal content, and RDB decay over 0 to 240 min; EP2473281B1 is more claim- and concept-focused in the extracted evidence.
- The metathesis-only patents (US7919563B2, US7772328B2) use co-olefins such as ethylene and 1-hexene at 80°C, with catalyst loadings from 0.05 to 0.25 phr and ethylene pressure of 6.9×10^5 Pa or 500 psi, whereas the simultaneous hydrogenation patents emphasize hydrogen atmosphere and catalyst identity over detailed pressure/time windows in the extracted text.
- US7846995B2 is a downstream purification patent rather than a synthesis patent; it complements the synthesis patents by addressing Ru residue removal from optionally hydrogenated nitrile rubber using ion-exchange resins.
- US10759885B2 and US10717906B2 are not NBR patents, but they show broader hydrogenation trends in elastomers and resins: hydrogenation is used to improve processability, aging resistance, and ozone stability in related polymer systems.
- Across the HNBR-focused patents, the dominant process trend is moving from two-step metathesis then hydrogenation toward one-step simultaneous metathesis/hydrogenation, with catalyst selection and residue control becoming central to manufacturability.

4. CONCLUSION

The patent evidence supports two practical synthesis directions for hydrogenated NBR: controlled metathesis of nitrile rubber to lower molecular weight, and single-step simultaneous metathesis/hydrogenation to generate low-RDB HNBR more efficiently. The strongest HNBR-specific evidence favors Ru/Os catalyst systems, optional co-olefins, and hydrogen atmosphere, with target product windows around Mw 20,000 to 250,000 and Mooney viscosity from about 1 to 50. For process development, the most evidence-backed parameters are moderate temperatures near 80°C, catalyst loadings in the 0.05 to 0.25 phr range for metathesis-based depolymerization, and careful control of hydrogenation catalyst choice to avoid excessive nitrile reduction. Post-reaction purification with functionalized ion-exchange resins is also important to reduce residual ruthenium. Overall, the best-supported route for HNBR synthesis is a Ru-catalyzed one-step hydrogenation/metathesis process, followed by metal-removal cleanup, when the goal is low residual unsaturation, lower viscosity, and improved processability.

5. REFERENCES

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- US10759885B2 | Hydrogenated branched conjugated diene copolymer, rubber composition and ... | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10759885B2/en>
- US10717906B2 | Self-adhesive compound and use thereof in an adhesive tape | Not disclosed | US | 2020 | <https://patents.google.com/patent/US10717906B2/en>